Phase 0 · Classical Mechanics — Lesson 3: Noether's Theorem

Self-study companion to *Quantum Field Theory for the Gifted Amateur*

Lesson overview

Prerequisites. Lesson 1 (action, Euler–Lagrange equations for particles and fields, the Klein–Gordon Lagrangian) and Lesson 2 (Legendre transform, Hamilton's equations, Poisson brackets $\{q,\, p\} = +1$ with $\dot{A} = \{A,\, H\} + \partial_t A$, field momentum $\pi = \partial\mathcal{L}/\partial\dot{\phi}$).
Time estimate. About 2–3 hours, plus the exercises.
Learning goals. By the end you should be able to:

- prove Noether's theorem for particles and for fields, from scratch;
- produce the conserved charge for translations, rotations, time translation and boosts, and know the Hamiltonian meaning of each ("the charge generates the symmetry");
- derive the $U(1)$ current of the complex scalar field and the canonical energy–momentum tensor $T^{\mu\nu}$, and check $T^{00} = \mathcal{H}$;
- explain, qualitatively, how promoting a global symmetry to a local one forces a gauge field.

Conventions throughout: $\hbar = c = 1$, metric $(+,-,-,-)$, $\partial_\mu = \partial/\partial x^\mu$, so $\partial^\mu = (\partial_t, -\nabla)$. Lancaster & Blundell is abbreviated L&B.

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1 Motivation: symmetry and conservation

In classical mechanics you learned three conservation laws (energy, momentum, angular momentum) and probably learned them as three separate facts, each proved by a separate short calculation. Noether's theorem (Emmy Noether, 1918) says they are three instances of a single principle:

> *Every continuous symmetry of the action gives a conserved quantity.*

Time-translation invariance gives energy; space-translation invariance gives momentum; rotation invariance gives angular momentum. The mapping is mechanical, and it is a *recipe*: you hand it a symmetry and it hands you the conserved charge.

This matters enormously in QFT, where symmetry organizes almost everything:

- Electric charge (and baryon number, lepton number, . . .) is the Noether charge of an internal $U(1)$ phase symmetry.
- Energy and momentum of a field are the Noether charges of spacetime translations, and they are packaged into the *energy–momentum tensor* $T^{\mu\nu}$, the object that sources gravity.
- Gauge theory—electromagnetism, the weak and strong forces—starts by asking what happens if a symmetry is imposed *locally*.

In a field theory there are infinitely many degrees of freedom and conservation laws must be *local*: charge does not just stay constant, it cannot jump from one place to another; it flows continuously. That is the continuity equation $\partial_\mu J^\mu = 0$, which we will derive.

The plan: first the clean mechanics of a finite number of particles (where every step is transparent), then the Hamiltonian reading (charges generate symmetries), then the field-theory version, then the key examples that L&B uses.

2 Warm-up: cyclic coordinates

Recall from Lesson 1 the Euler–Lagrange equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0. \tag{2.1}$$

Define the conjugate momentum $p_i = \partial L/\partial \dot{q}_i$. Then the equation reads $\dot{p}_i = \partial L/\partial q_i$. So:

Key result: cyclic coordinates

If L does not depend on the coordinate q_k (it is *cyclic* or *ignorable*), then $p_k = \partial L/\partial \dot{q}_k$ is conserved.

Notice what the hypothesis says: L is unchanged under the shift $q_k \to q_k + \varepsilon$. That is a symmetry, and this is the simplest case of Noether's theorem.

Worked example 2.1 (Free particle)**.** $L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$ does not contain x, y, z. Hence $p_x = m\dot{x}$, $p_y = m\dot{y}$, $p_z = m\dot{z}$ are conserved: translation invariance $\Rightarrow$ momentum conservation.

Worked example 2.2 (Central potential in the plane)**.** For $V = V(r)$ in polar coordinates, $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\varphi}^2) - V(r)$. The angle φ does not appear, so

$$p_\varphi = \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi} = L_z \tag{2.2}$$

is conserved. This is the z-component of angular momentum $\boldsymbol{r} \times \boldsymbol{p}$: rotation invariance $\Rightarrow$ angular momentum conservation. (In Cartesian coordinates the symmetry $x \to x - \varepsilon y$, $y \to y + \varepsilon x$ is far from obvious as a "cyclic coordinate"; the polar coordinate hid it in plain sight. Noether's theorem removes the need to hunt for good coordinates.)

The limitation is clear: sometimes the symmetry is not a simple shift of a single coordinate (a rotation in Cartesian coordinates; a boost; a time translation). We need the general statement.

3 Noether's theorem for particles

3.1 Infinitesimal symmetries

Consider a mechanical system with coordinates $q_i(t)$, $i = 1, \dots, N$ and Lagrangian $L(q, \dot{q}, t)$. An *infinitesimal transformation* is

$$q_i(t) \;\longrightarrow\; q_i(t) + \delta q_i(t), \qquad \delta q_i = \varepsilon\, K_i(q, t), \tag{3.1}$$

with ε an infinitesimal constant parameter and K_i a given function (it may depend on q, and even on t or $\dot{q}$) that says *how* the coordinates are moved. The transformation is applied to an entire path, $q_i(t)$ is replaced by $q_i(t) + \varepsilon K_i(q(t), t)$ at each time. Because ε is constant, the velocities change by

$$\delta\dot{q}_i = \frac{\mathrm{d}}{\mathrm{d}t}\delta q_i = \varepsilon\,\dot{K}_i, \tag{3.2}$$

where $\dot{K}_i = \mathrm{d}K_i/\mathrm{d}t$ is the *total* time derivative along the path. We work to first order in ε throughout.

Definition 3.1 (Symmetry)**.** The transformation (3.1) is a *symmetry* if the Lagrangian changes only by a total time derivative:

$$\delta L = \varepsilon \frac{\mathrm{d}F}{\mathrm{d}t} \tag{3.3}$$

for some function $F(q,t)$, for *arbitrary* paths $q(t)$ (not only solutions). The *strict* case is $\delta L = 0$, i.e. $F = 0$.

Why is a total derivative allowed? The action is $S = \int L\,\mathrm{d}t$, so $\delta S = \varepsilon[F]_{t_1}^{t_2}$ is a pure boundary term, which does not affect the equations of motion (recall Lesson 1: L and $L + \mathrm{d}G/\mathrm{d}t$ describe the same physics). So (3.3) is the right notion: the symmetry maps solutions to solutions. You will meet important cases in which $F \neq 0$ (Galilean boosts below).

3.2 The theorem and its proof

Theorem 3.1 (Noether, particles)**.** *If $\delta q_i = \varepsilon K_i$ is a symmetry in the sense of (3.3), then*

$$Q = \sum_i \frac{\partial L}{\partial \dot{q}_i} K_i - F = \sum_i p_i K_i - F \tag{3.4}$$

is conserved along every solution of the equations of motion: $\mathrm{d}Q/\mathrm{d}t = 0$.

Proof. Compute δL in two ways. First, by the chain rule, using (3.1) and (3.2) (there is no variation of t, so $\partial L/\partial t$ does not contribute):

$$\delta L = \sum_i \left(\frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\delta \dot{q}_i \right) = \varepsilon \sum_i \left(\frac{\partial L}{\partial q_i} K_i + p_i \dot{K}_i \right). \tag{3.5}$$

Now go *on-shell*, i.e. assume $q_i(t)$ solves the Euler–Lagrange equations, so that $\partial L/\partial q_i = \dot{p}_i$:

$$\delta L = \varepsilon \sum_i \left(\dot{p}_i K_i + p_i \dot{K}_i \right) = \varepsilon \frac{\mathrm{d}}{\mathrm{d}t}\Big(\sum_i p_i K_i \Big). \tag{3.6}$$

Second, by the symmetry assumption, $\delta L = \varepsilon\,\mathrm{d}F/\mathrm{d}t$. Equating the two:

$$\varepsilon \frac{\mathrm{d}}{\mathrm{d}t}\Big(\sum_i p_i K_i - F \Big) = 0. \tag{3.7}$$

□

The structure of the proof is worth internalizing: the *off-shell* statement "$\delta L = \varepsilon \dot{F}$" is a property of the Lagrangian (the symmetry), while the *on-shell* chain-rule computation converts it into a statement about motion. Conservation holds only on solutions.

Key result: Noether's theorem (particles)

Symmetry $\delta q_i = \varepsilon K_i$, $\delta L = \varepsilon\,\mathrm{d}F/\mathrm{d}t \Longrightarrow Q = \sum_i \dfrac{\partial L}{\partial \dot{q}_i} K_i - F$ is conserved.

Watch out

[Sign conventions for ε] Whether you write $q \to q+\varepsilon K$ or $q \to q-\varepsilon K$ flips the sign of K and of Q, but conservation is unaffected. Also Q is defined only up to an overall constant factor (rescale ε) and up to adding a constant. Do not worry about overall signs when checking conservation; do worry about them when you need Q to equal a *specific* physical quantity such as $+\boldsymbol{p}$ or $+H$, or when you later compare conventions between books (L&B, Tong, Peskin, ...).

3.3 Worked examples

Worked example 3.1 (Spatial translation ⇒ momentum)**.** Take N particles with $L = \sum_a \frac{1}{2} m_a \dot{\boldsymbol{r}}_a^2 - V(\boldsymbol{r}_a - \boldsymbol{r}_b)$ and shift every particle by the same vector: $\delta \boldsymbol{r}_a = \varepsilon \boldsymbol{n}$ for a fixed unit vector $\boldsymbol{n}$. So $K_a = \boldsymbol{n}$ for every particle. Since V depends only on differences, $\delta L = 0$ and $F = 0$. Then

$$Q = \sum_a \boldsymbol{p}_a \cdot \boldsymbol{n} = \boldsymbol{n} \cdot \boldsymbol{P}, \tag{3.8}$$

the component of the total momentum along $\boldsymbol{n}$. If the potential also depends on a fixed external source (e.g. $V(\boldsymbol{r})$ not shift-invariant in some direction), the symmetry survives only for those directions $\boldsymbol{n}$ along which V is constant.

Worked example 3.2 (Rotation ⇒ angular momentum)**.** Rotate every particle about the axis $\boldsymbol{n}$ ($|\boldsymbol{n}| = 1$) through the small angle ε. A rotation by small angle ε about $\boldsymbol{n}$ moves a position vector as

$$\delta \boldsymbol{r}_a = \varepsilon\, \boldsymbol{n} \times \boldsymbol{r}_a, \qquad K_a = \boldsymbol{n} \times \boldsymbol{r}_a. \tag{3.9}$$

(Check: for $\boldsymbol{n} = \hat{z}$, $\boldsymbol{r} = (x, y, z)$ one gets $\delta \boldsymbol{r} = \varepsilon(-y, x, 0)$, which is the counter-clockwise rotation $x \to x\cos\varepsilon - y\sin\varepsilon \approx x - \varepsilon y$, $y \to y + \varepsilon x$.) For a Lagrangian built from scalar products, $\dot{\boldsymbol{r}}_a^2$, $\boldsymbol{r}_a \cdot \boldsymbol{r}_b$, $|\boldsymbol{r}_a - \boldsymbol{r}_b|$, which are rotation invariants, $\delta L = 0$. Verify the kinetic term: $\delta(\dot{\boldsymbol{r}}^2) = 2\dot{\boldsymbol{r}} \cdot (\varepsilon\, \boldsymbol{n} \times \dot{\boldsymbol{r}}) = 0$ because $\dot{\boldsymbol{r}} \perp (\boldsymbol{n} \times \dot{\boldsymbol{r}})$. Then

$$Q = \sum_a \boldsymbol{p}_a \cdot (\boldsymbol{n} \times \boldsymbol{r}_a) = \sum_a \boldsymbol{n} \cdot (\boldsymbol{r}_a \times \boldsymbol{p}_a) = \boldsymbol{n} \cdot \boldsymbol{J}, \tag{3.10}$$

where we used the cyclic property of the triple product $\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b} \cdot (\boldsymbol{c} \times \boldsymbol{a})$. Since this holds for each $\boldsymbol{n}$, all three components of $\boldsymbol{J} = \sum_a \boldsymbol{r}_a \times \boldsymbol{p}_a$ are conserved *if the system is invariant under rotations about all three axes.* For a single particle in a central potential $V(|\boldsymbol{r}|)$ this is true, which is the origin of conservation of the vector $\boldsymbol{L}$ (and hence planar motion). For an axially-symmetric potential only L_z survives.

Worked example 3.3 (Time translation ⇒ energy)**.** This one does not look like (3.1), because it moves the system in *time* rather than in configuration space. But shifting the time origin, $t \to t+\varepsilon$, acts on a given path as

$$q_i(t) \to q_i(t+\varepsilon) = q_i(t) + \varepsilon \dot{q}_i(t), \qquad \text{i.e.} \quad K_i = \dot{q}_i. \tag{3.11}$$

(Here K_i depends on the velocity; the proof of Theorem 3.1 did not care, since only $\delta \dot{q}_i = \frac{\mathrm{d}}{\mathrm{d}t} \delta q_i$ was used.) Now suppose $L = L(q, \dot{q})$ has no explicit time dependence. Then the change in L under this relabelling is simply the change of $L(q(t), \dot{q}(t))$ under $t \to t + \varepsilon$:

$$\delta L = \sum_i \left(\frac{\partial L}{\partial q_i} \varepsilon \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \varepsilon \ddot{q}_i \right) = \varepsilon \frac{\mathrm{d}L}{\mathrm{d}t}, \tag{3.12}$$

where the last equality uses $\partial L/\partial t = 0$. So $F = L$. (Here F depends on $\dot{q}$ as well as q; that is harmless, because the proof only uses $\delta L = \varepsilon\, \mathrm{d}F/\mathrm{d}t$ along paths, never the form of F.) The conserved charge is

$$Q = \sum_i p_i \dot{q}_i - L = H, \tag{3.13}$$

the Hamiltonian (Lesson 2): conservation of energy. If instead $\partial L/\partial t \neq 0$ then $\delta L = \varepsilon(\mathrm{d}L/\mathrm{d}t - \partial_t L)$ is not a total derivative, there is no symmetry, and H is not conserved; Exercise 9 makes this precise.

Worked example 3.4 (Galilean boost ⇒ centre-of-mass motion; a symmetry with $F \neq 0$)**.** Take the same N-particle system, $L = \sum_a \frac{1}{2} m_a \dot{\boldsymbol{r}}_a^2 - V(\boldsymbol{r}_a - \boldsymbol{r}_b)$, and change to a frame moving with small velocity $\varepsilon \boldsymbol{n}$:

$$\delta \boldsymbol{r}_a = \varepsilon\, \boldsymbol{n}\, t, \qquad K_a = \boldsymbol{n}\, t \quad \text{(explicitly time dependent)}, \qquad \delta \dot{\boldsymbol{r}}_a = \varepsilon \boldsymbol{n}. \tag{3.14}$$

The potential depends on differences and is unchanged. The kinetic term changes:

$$\delta L = \sum_a m_a \dot{\boldsymbol{r}}_a \cdot \varepsilon \boldsymbol{n} = \varepsilon \frac{\mathrm{d}}{\mathrm{d}t}\Big(\boldsymbol{n} \cdot \sum_a m_a \boldsymbol{r}_a\Big) = \varepsilon \frac{\mathrm{d}}{\mathrm{d}t}(M\, \boldsymbol{n} \cdot \boldsymbol{R}), \tag{3.15}$$

with $M = \sum m_a$ and $\boldsymbol{R} = \sum m_a \boldsymbol{r}_a / M$ the centre of mass. This is nonzero but a total derivative: $F = M \boldsymbol{n} \cdot \boldsymbol{R}$. The charge is

$$Q = \sum_a \boldsymbol{p}_a \cdot \boldsymbol{n}\, t - M \boldsymbol{n} \cdot \boldsymbol{R} = \boldsymbol{n} \cdot (\boldsymbol{P}\, t - M \boldsymbol{R}). \tag{3.16}$$

Conservation of Q for all $\boldsymbol{n}$ says $M\boldsymbol{R} - \boldsymbol{P}t = \text{const}$, i.e.

$$\boldsymbol{R}(t) = \boldsymbol{R}(0) + \frac{\boldsymbol{P}}{M}\, t, \tag{3.17}$$

the centre of mass moves uniformly. Notice that this is a genuinely new piece of information (it is *not* just momentum conservation; it is the statement that $\dot{\boldsymbol{R}} = \boldsymbol{P}/M$ is constant, which is Newton's second law summed over particles), and that it only appeared because we allowed $F \neq 0$ and t-dependent K.

Watch out

[Forgetting F] If you insisted on $\delta L = 0$ you would conclude the boost is not a symmetry, and miss the conservation law. Always compute δL and ask whether it is a total derivative. Conversely, don't be fooled by a non-vanishing δL: it is only a problem if it is *not* a total derivative.

4 The Hamiltonian view: charges generate symmetries

Recall from Lesson 2 that for any function $A(q, p, t)$,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \{A,\, H\} + \frac{\partial A}{\partial t}, \qquad \{A,\, B\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right), \tag{4.1}$$

so $\{q_i,\, p_j\} = +\delta_{ij}$. A quantity $Q(q, p)$ with no explicit time dependence is therefore conserved if and only if

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \{Q,\, H\} = 0. \tag{4.2}$$

Key result: charges generate their symmetries

The Noether charge Q (for a symmetry with no explicit t-dependence) generates the transformation it came from:

$$\delta q_i = \varepsilon \{q_i, Q\}, \qquad \delta p_i = \varepsilon \{p_i, Q\}. \tag{4.3}$$

Moreover, $\{Q, H\} = 0$ means precisely that H is invariant under this transformation: $\delta H = \varepsilon \{H, Q\} = -\varepsilon \{Q, H\} = 0$.

We verify the first statement in the two key cases (for the symmetries, $F = 0$, $Q = \sum_i p_i K_i$ with K independent of p, then $\{q_i, Q\} = K_i$ immediately, since $\{q_i, p_j\} = \delta_{ij}$; the content is in the check that it also does the right thing to p).

Worked example 4.1 (Momentum generates translations)**.** Let $Q = \boldsymbol{n} \cdot \boldsymbol{p}$. Then $\{x_i, \boldsymbol{n} \cdot \boldsymbol{p}\} = n_j \{x_i, p_j\} = n_i$, so $\delta \boldsymbol{r} = \varepsilon \boldsymbol{n}$. And $\{p_i, \boldsymbol{n} \cdot \boldsymbol{p}\} = 0$: momenta are untouched. Translation invariance of H means $\{H, \boldsymbol{n} \cdot \boldsymbol{p}\} = \boldsymbol{n} \cdot \nabla V = 0$ (for $H = \boldsymbol{p}^2/2m + V$), i.e. V does not change along $\boldsymbol{n}$, as expected.

Worked example 4.2 (Angular momentum generates rotations)**.** Let $Q = \boldsymbol{n} \cdot (\boldsymbol{r} \times \boldsymbol{p}) = \epsilon_{jkl} n_j x_k p_l$ (summation convention). Using $\{x_i, p_l\} = \delta_{il}$,

$$\{x_i, Q\} = \epsilon_{jkl} n_j x_k \delta_{il} = \epsilon_{jki} n_j x_k = \epsilon_{ijk} n_j x_k = (\boldsymbol{n} \times \boldsymbol{r})_i, \tag{4.4}$$

using the cyclic symmetry of ϵ. So $\delta \boldsymbol{r} = \varepsilon\, \boldsymbol{n} \times \boldsymbol{r}$: exactly the rotation of Section 3. For the momentum, using $\{p_i, x_k\} = -\delta_{ik}$,

$$\{p_i, Q\} = \epsilon_{jkl} n_j \{p_i, x_k\} p_l = -\epsilon_{jil} n_j p_l = \epsilon_{ijl} n_j p_l = (\boldsymbol{n} \times \boldsymbol{p})_i, \tag{4.5}$$

so $\delta \boldsymbol{p} = \varepsilon\, \boldsymbol{n} \times \boldsymbol{p}$: the momentum vector is rotated in the same way, as it must be for $\boldsymbol{r}$ and $\boldsymbol{p}$ to remain "vectors".

Bridge to QFT

This is one of the most important sentences of the whole course: *the Noether charge is the generator of its symmetry.* In quantum mechanics, $\{\cdot, \cdot\} \to -\mathrm{i}[\cdot, \cdot]$ (Dirac, Lesson 2), a conserved classical Q becomes a Hermitian operator $\hat{Q}$ with $[\hat{Q}, \hat{H}] = 0$, and the symmetry transformation is implemented by the *unitary* operator $U = \mathrm{e}^{\mathrm{i}\varepsilon \hat{Q}}$ (up to sign conventions), with $U \hat{H} U^\dagger = \hat{H}$. The eigenvalues of $\hat{Q}$ label states and are conserved in time. In L&B this appears in ch. 10 ("Symmetry") and again in ch. 12, where the $U(1)$ charge operator $\hat{Q}$ counts particles minus antiparticles, and wherever the translation generators become the momentum operator $\hat{\boldsymbol{P}}$ with $\hat{\boldsymbol{P}} |\boldsymbol{p}\rangle = \boldsymbol{p} |\boldsymbol{p}\rangle$.

5 Noether's theorem for fields

Now to fields $\phi_a(x)$, $x^\mu = (t, \boldsymbol{x})$, with action $S = \int \mathrm{d}^4 x\, \mathcal{L}(\phi_a, \partial_\mu \phi_a)$. From Lesson 1, the Euler–Lagrange equations are

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0. \tag{5.1}$$

Consider an *internal* transformation (one that rescales/rotates the field values at each point, leaving x alone):

$$\phi_a(x) \to \phi_a(x) + \varepsilon\, \Delta\phi_a(x), \tag{5.2}$$

where $\Delta\phi_a$ is a given function of the fields. Then $\delta(\partial_\mu \phi_a) = \varepsilon\, \partial_\mu(\Delta\phi_a)$.

The field analogue of (3.3) is that the Lagrangian density changes by a total *four*-divergence:

$$\delta\mathcal{L} = \varepsilon\, \partial_\mu W^\mu \tag{5.3}$$

for some $W^\mu(\phi)$, for arbitrary field configurations. This leaves the action invariant up to a surface term at the boundary of spacetime, which does not affect the equations of motion. The strict case is $W^\mu = 0$.

Theorem 5.1 (Noether, fields)**.** *If $\phi_a \to \phi_a + \varepsilon\Delta\phi_a$ satisfies (5.3), then*

$$J^\mu = \sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \Delta\phi_a - W^\mu \tag{5.4}$$

obeys $\partial_\mu J^\mu = 0$ on solutions of (5.1).

Proof. By the chain rule,

$$\delta\mathcal{L} = \varepsilon \sum_a \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \Delta\phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\mu(\Delta\phi_a) \right]. \tag{5.5}$$

On-shell, (5.1) gives $\partial\mathcal{L}/\partial\phi_a = \partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi_a))$. Substituting, the two terms combine via the product rule into a divergence:

$$\delta\mathcal{L} = \varepsilon\, \partial_\mu \Big[\sum_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \Delta\phi_a \Big]. \tag{5.6}$$

Setting this equal to $\varepsilon\, \partial_\mu W^\mu$ from (5.3) gives $\partial_\mu J^\mu = 0$. □

The conserved charge. Split the divergence: $\partial_\mu J^\mu = \partial_t J^0 + \nabla \cdot \boldsymbol{J} = 0$, where $\boldsymbol{J} = (J^1, J^2, J^3)$. Define

$$Q = \int \mathrm{d}^3x\, J^0. \tag{5.7}$$

Then

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \int \mathrm{d}^3x\, \partial_t J^0 = -\int \mathrm{d}^3x\, \nabla \cdot \boldsymbol{J} = -\oint_{S_\infty} \boldsymbol{J} \cdot \mathrm{d}\boldsymbol{S} = 0, \tag{5.8}$$

by the divergence theorem, provided the fields (hence $\boldsymbol{J}$) fall off fast enough at spatial infinity that the surface integral vanishes. (More precisely, $|\boldsymbol{J}|$ must decay faster than $1/r^2$; this holds for configurations of finite energy in all examples here.)

Key result: Noether's theorem (fields)

$\Delta\phi_a$ with $\delta\mathcal{L} = \varepsilon\, \partial_\mu W^\mu \Rightarrow$ current $J^\mu = \sum_a \dfrac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \Delta\phi_a - W^\mu$, $\partial_\mu J^\mu = 0$, and charge $Q = \int \mathrm{d}^3x\, J^0$ constant in time.

Compare with the particle case: $\partial\mathcal{L}/\partial(\partial_0\phi_a) = \pi_a$ is the field momentum, so $J^0 = \sum_a \pi_a \Delta\phi_a - W^0$ is exactly "$\sum p_i K_i - F$" density.

Watch out

[On-shell only] $\partial_\mu J^\mu = 0$ holds only when the fields satisfy their equations of motion. Off-shell, $\partial_\mu J^\mu \neq 0$ in general (in the proof, the Euler–Lagrange equation was used). This is exactly like particle mechanics, where Q is constant only along true trajectories. In the quantum theory this is the difference between "Ward identities" and operator statements, but for now simply remember to use the field equation when you check conservation.

Watch out

[Currents are not unique] Adding to J^μ a term $\partial_\nu A^{[\mu\nu]}$ with $A^{[\mu\nu]} = -A^{[\nu\mu]}$ leaves $\partial_\mu J^\mu = 0$ intact (because $\partial_\mu \partial_\nu A^{[\mu\nu]} = 0$) and does not change Q (the extra piece in J^0 is a divergence $\partial_i A^{[0i]}$ that integrates to a surface term). So the current is defined up to such terms, and it is Noether's recipe that selects one representative.

Worked example 5.1 (Continuity equation)**.** For a fluid with density ρ and velocity $\boldsymbol{v}$, $J^\mu = (\rho, \rho\boldsymbol{v})$ obeys $\partial_\mu J^\mu = \partial_t \rho + \nabla \cdot (\rho \boldsymbol{v}) = 0$, the familiar continuity equation: mass cannot vanish here and appear there; it must flow through the intervening space. That is the content of a *local* conservation law. Exercise 8 makes the local–global connection precise.

6 Key field examples

6.1 The complex scalar field and $U(1)$ charge

Take a complex scalar field $\psi(x)$ with

$$\mathcal{L} = \partial_\mu \psi^* \partial^\mu \psi - m^2 \psi^* \psi. \tag{6.1}$$

As in Lesson 1, treat ψ and ψ^* as independent fields. Varying ψ^* gives $(\partial_\mu \partial^\mu + m^2)\psi = 0$ and varying ψ gives the conjugate equation: both obey Klein–Gordon.

The Lagrangian depends on ψ, ψ^* only through $\psi^*\psi$ and derivative combinations $\partial\psi^* \partial\psi$, so it is invariant under the global phase rotation

$$\psi \to \mathrm{e}^{-\mathrm{i}\alpha}\psi, \qquad \psi^* \to \mathrm{e}^{+\mathrm{i}\alpha}\psi^*, \qquad \alpha \in \mathbb{R} \text{ constant}. \tag{6.2}$$

(The group of phases is $U(1)$.) We have chosen the sign $\mathrm{e}^{-\mathrm{i}\alpha}$ for a reason explained below. Infinitesimally, with $\alpha = \varepsilon$:

$$\Delta\psi = -\mathrm{i}\psi, \qquad \Delta\psi^* = +\mathrm{i}\psi^*. \tag{6.3}$$

Because $\psi^*\psi \to \psi^*\psi$ and $\partial_\mu\psi^*\partial^\mu\psi \to \partial_\mu\psi^*\partial^\mu\psi$ (the phase is the same everywhere in spacetime, so it passes through derivatives), $\delta\mathcal{L} = 0$ and $W^\mu = 0$. Theorem 5.1, summing over the two independent fields ψ, ψ^*:

$$\begin{aligned} J^\mu &= \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)}\Delta\psi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi^*)}\Delta\psi^* \\ &= \partial^\mu\psi^* \,(-\mathrm{i}\psi) + \partial^\mu\psi \,(+\mathrm{i}\psi^*) \\ &= \mathrm{i}\big(\psi^* \partial^\mu \psi - \psi\, \partial^\mu \psi^*\big). \end{aligned} \tag{6.4}$$

Here $\partial\mathcal{L}/\partial(\partial_\mu\psi) = \partial^\mu\psi^*$ since $\mathcal{L} \ni \partial_\nu\psi^*\partial^\nu\psi$.

Let us verify $\partial_\mu J^\mu = 0$ directly (using the equations of motion):

$$\partial_\mu J^\mu = \mathrm{i}(\partial_\mu\psi^*\partial^\mu\psi + \psi^*\Box\psi - \partial_\mu\psi\partial^\mu\psi^* - \psi\Box\psi^*) = \mathrm{i}(\psi^*\Box\psi - \psi\Box\psi^*) = \mathrm{i}(-m^2\psi^*\psi + m^2\psi\psi^*) = 0, \tag{6.5}$$

with $\Box = \partial_\mu\partial^\mu$ and $\Box\psi = -m^2\psi$. Note this required the field equations (Pitfall above).

Key result: $U(1)$ current of the complex scalar field

With $\psi \to \mathrm{e}^{-\mathrm{i}\alpha}\psi$:

$$J^\mu = \mathrm{i}(\psi^*\partial^\mu\psi - \psi\,\partial^\mu\psi^*), \qquad Q = \int \mathrm{d}^3x\, J^0 = \mathrm{i}\int \mathrm{d}^3x\,(\psi^*\dot\psi - \psi\dot\psi^*). \tag{6.6}$$

Watch out

[Overall sign and normalization are conventions] Had we used $\psi \to \mathrm{e}^{+\mathrm{i}\alpha}\psi$ (i.e. $\Delta\psi = +\mathrm{i}\psi$) the same derivation would give $J^\mu = \mathrm{i}(\psi\partial^\mu\psi^* - \psi^*\partial^\mu\psi) = -J^\mu_{\text{above}}$. Different books pick different signs (and some insert factors of the charge e or a $1/2$). The sign matters physically only in that it decides whether *particle* or *antiparticle* carries $Q = +1$. Our choice is the one in which a positive-frequency mode carries positive charge, as shown next. When you reach L&B ch. 12, **check their stated convention** for J^μ and for the transformation $\psi \to \mathrm{e}^{\pm\mathrm{i}\alpha}\psi$ and adjust signs accordingly.

Interpretation. The density is $J^0 = \mathrm{i}(\psi^*\dot\psi - \psi\dot\psi^*)$. For a single positive-frequency plane wave $\psi = A\,\mathrm{e}^{-\mathrm{i}Et+\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{x}}$, $\dot\psi = -\mathrm{i}E\psi$ and $\dot\psi^* = +\mathrm{i}E\psi^*$, so

$$J^0 = \mathrm{i}(-\mathrm{i}E|\psi|^2 - \mathrm{i}E|\psi|^2) = 2E|A|^2 > 0. \tag{6.7}$$

Likewise $J^i = \mathrm{i}(\psi^*\partial^i\psi - \psi\partial^i\psi^*) = \mathrm{i}(-\psi^*\partial_i\psi + \psi\partial_i\psi^*) = 2p_i|A|^2$, so $\boldsymbol{J} = 2\boldsymbol{p}|A|^2 = (\boldsymbol{p}/E)J^0$, which is $\rho\boldsymbol{v}$ with $\boldsymbol{v}$ the group velocity, as it should be. A negative-frequency mode $\mathrm{e}^{+\mathrm{i}Et}$ gives $J^0 < 0$. Thus Q counts (number of positive-frequency excitations) $-$ (number of negative-frequency excitations): once quantized, the field has particles of charge $+1$ and antiparticles of charge -1. Note that J^0 is *not* positive definite in general, so it cannot be interpreted as a probability density as in non-relativistic QM—a historical headache for the Klein–Gordon equation, resolved by QFT.

In Hamiltonian language, $\pi = \partial\mathcal{L}/\partial\dot\psi = \dot\psi^*$ and $\pi^* = \dot\psi$, so $Q = \mathrm{i}\int \mathrm{d}^3x\,(\psi^*\pi^* - \psi\pi)$; Exercise 11 checks that Q generates the phase rotation via $\delta\psi = \varepsilon\,\{\psi, Q\}$.

Bridge to QFT

This is L&B ch. 12 (the complex scalar field). There $\hat\psi$ contains annihilation operators $\hat a_{\boldsymbol{p}}$ for particles and creation operators $\hat b^\dagger_{\boldsymbol{p}}$ for antiparticles, and the Noether charge $\hat Q$ becomes (up to normal ordering) $\sum_{\boldsymbol{p}}(\hat a^\dagger_{\boldsymbol{p}}\hat a_{\boldsymbol{p}} - \hat b^\dagger_{\boldsymbol{p}}\hat b_{\boldsymbol{p}})$, number of particles minus antiparticles. That $[\hat Q, \hat H] = 0$ means this difference is conserved in every interaction that respects the symmetry; e.g. particle–antiparticle pairs are created together. Ch. 10 discusses the same structure for a general internal symmetry.

6.2 Spacetime translations and the energy–momentum tensor

Now consider the symmetry that is not internal: translations of spacetime,

$$x^\mu \to x^\mu + \varepsilon^\mu \quad \Longrightarrow \quad \phi(x) \to \phi(x+\varepsilon) = \phi(x) + \varepsilon^\nu \partial_\nu \phi(x). \tag{6.8}$$

(The sign: we shift the *argument*, as with $q(t+\varepsilon)$ in the time-translation example. The fields are evaluated at the displaced point.) So $\Delta_{(\nu)}\phi = \partial_\nu \phi$, one transformation for each of the four values of ν (take ε^ν nonzero only for one ν at a time). If $\mathcal{L}$ has no explicit x-dependence (the Lagrangian density is built only from ϕ and $\partial\phi$), then $\mathcal{L}$ is just a scalar function evaluated at the new point, so

$$\delta\mathcal{L} = \varepsilon^\nu \partial_\nu \mathcal{L} = \varepsilon^\nu\, \partial_\mu(\delta^\mu{}_\nu \mathcal{L}), \qquad \text{so} \qquad W^\mu_{(\nu)} = \delta^\mu{}_\nu \mathcal{L}. \tag{6.9}$$

Theorem 5.1 gives four conserved currents, one for each ν:

$$(J^\mu)_{\ \nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\, \partial_\nu \phi - \delta^\mu{}_\nu \mathcal{L}. \tag{6.10}$$

Raising the index ν with the metric defines the (canonical) energy–momentum tensor:

Key result: canonical energy–momentum tensor

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\, \partial^\nu \phi - g^{\mu\nu}\mathcal{L}, \qquad \partial_\mu T^{\mu\nu} = 0. \tag{6.11}$$

(For several fields, sum the first term over fields.) For each fixed ν, $P^\nu = \int \mathrm{d}^3x\, T^{0\nu}$ is conserved.

Worked example 6.1 (Real Klein–Gordon field)**.** $\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\dot\phi^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2$. Here $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$, so

$$T^{\mu\nu} = \partial^\mu\phi\, \partial^\nu\phi - g^{\mu\nu}\mathcal{L}. \tag{6.12}$$

Energy density. $\mu = \nu = 0$, $g^{00} = 1$:

$$T^{00} = \dot\phi^2 - \mathcal{L} = \dot\phi^2 - \tfrac{1}{2}\dot\phi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 = \tfrac{1}{2}\dot\phi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2. \tag{6.13}$$

With $\pi = \partial\mathcal{L}/\partial\dot\phi = \dot\phi$ this is $\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 = \mathcal{H}$, the Hamiltonian density from Lesson 2. In general $T^{00} = \pi\dot\phi - \mathcal{L} = \mathcal{H}$, the field version of $H = p\dot q - L$. Thus $P^0 = \int \mathrm{d}^3x\, T^{00} = H$, the energy, is conserved.

Momentum density. $\mu = 0$, $\nu = i$, with $g^{0i} = 0$:

$$T^{0i} = \dot\phi\, \partial^i\phi = -\dot\phi\, \partial_i\phi = -\pi\, \partial_i\phi. \tag{6.14}$$

So the conserved field momentum is

$$P^i = \int \mathrm{d}^3x\, T^{0i} = -\int \mathrm{d}^3x\, \pi\, \partial_i\phi, \qquad \text{i.e.} \quad \boldsymbol{P} = -\int \mathrm{d}^3x\, \pi\, \nabla\phi. \tag{6.15}$$

Sanity check, via the Hamiltonian view of Section 4: with $\{\phi(\boldsymbol{x}), \pi(\boldsymbol{y})\} = \delta^3(\boldsymbol{x}-\boldsymbol{y})$ one finds $\{\phi(\boldsymbol{x}), \boldsymbol{P}\} = -\nabla\phi(\boldsymbol{x})$, so $\delta\phi = \varepsilon\, \{\phi, \boldsymbol{n}\cdot\boldsymbol{P}\} = -\varepsilon\, \boldsymbol{n}\cdot\nabla\phi = \phi(\boldsymbol{x} - \varepsilon\boldsymbol{n}) - \phi(\boldsymbol{x})$: $\boldsymbol{P}$ moves the field profile *forward* by $\varepsilon\boldsymbol{n}$, as a momentum should. This does not contradict the transformation $\phi(x+\varepsilon)$ used to derive $T^{\mu\nu}$: the Noether charge for $\Delta\phi = \partial_i\phi$ is $\int \pi\, \partial_i\phi = -P^i$, because $\partial_i = -\partial^i$; the sign flip is the usual $P_i = -P^i$ bookkeeping.

Watch out

[Index placement in $T^{\mu\nu}$] The first index μ is the *current* index: $\partial_\mu T^{\mu\nu} = 0$ is a conservation law, and $T^{0\nu}$ is the density of the charge P^ν. The second index ν labels *which* charge. With $\nu = 0$ you get energy, with $\nu = i$ momentum P^i. Note T^{0i} (momentum density) and T^{i0} (energy flux) agree for the scalar field but need not in general. Also, $\partial^\nu\phi$ has an upper index: $\partial^i = -\partial_i$, which is the source of the minus sign in $\boldsymbol{P} = -\int \pi\nabla\phi$.

Watch out

[Canonical $T^{\mu\nu}$ is not unique and not always symmetric] For the scalar field $T^{\mu\nu}$ is symmetric, but for fields with spin (e.g. the electromagnetic field) the canonical tensor in general is not, and it is not gauge-invariant for Maxwell theory. One can add $\partial_\lambda B^{\lambda\mu\nu}$ with $B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$ without spoiling $\partial_\mu T^{\mu\nu} = 0$ or changing P^ν (Belinfante's procedure produces a symmetric, gauge-invariant version, which is the one coupled to gravity). A massless scalar can also be "improved" to make it traceless (Exercise 6).

Bridge to QFT

In L&B (ch. 10, and again in ch. 11 when the Klein–Gordon field is quantized) the total momentum operator $\hat{\boldsymbol{P}} = -\int \mathrm{d}^3x\, \hat\pi\, \nabla\hat\phi$ and the Hamiltonian $\hat H = \int \mathcal{H}\, \mathrm{d}^3x$ are the generators of translations; in terms of mode operators they become $\sum_{\boldsymbol{p}} \boldsymbol{p}\, \hat a^\dagger_{\boldsymbol{p}}\hat a_{\boldsymbol{p}}$ and $\sum_{\boldsymbol{p}} E_{\boldsymbol{p}}\hat a^\dagger_{\boldsymbol{p}}\hat a_{\boldsymbol{p}}$. Seeing $T^{00} = \mathcal{H}$ here is the reason the Hamiltonian you build from the Legendre transform is "the" energy.

6.3 Lorentz transformations (statement only)

Lorentz transformations $x^\mu \to \Lambda^\mu{}_\nu x^\nu$ also leave a relativistic action invariant. Applying Noether's theorem (with the transformation now acting on both the argument and, for fields with spin, the field components) yields a rank-3 current $\mathcal{M}^{\lambda\mu\nu}$ antisymmetric in $\mu\nu$, with six conserved charges $M^{\mu\nu} = \int \mathrm{d}^3x\, \mathcal{M}^{0\mu\nu}$. Three of them ($M^{ij}$) are the components of angular momentum, which for a scalar field has the form $\boldsymbol{J} = \int \mathrm{d}^3x\, \boldsymbol{x} \times$ (momentum density), i.e. $J^{ij} = \int \mathrm{d}^3x\, (x^i T^{0j} - x^j T^{0i})$. The other three ($M^{0i}$) are the *boost charges*, the relativistic counterpart of the Galilean combination $\boldsymbol{P}t - M\boldsymbol{R}$ above: they are $K^i = \int \mathrm{d}^3x\, (x^i T^{00} - tT^{0i})$, expressing that the energy centroid moves with velocity $\boldsymbol{P}/E$.

Bridge to QFT

The full set of Lorentz charges (spin plus orbital angular momentum) returns when L&B reach spinors and the Dirac field (Part IX, from ch. 36); at this stage all you need is that $T^{\mu\nu}$ and the $U(1)$ current are the two conserved objects used repeatedly in the early chapters.

7 Local versus global symmetry: a preview

The phase rotation above used a *constant* α (a *global* symmetry): the same rotation everywhere in spacetime. What if we demand invariance under $\psi \to \mathrm{e}^{\mathrm{i}\alpha(x)}\psi$ with α depending on position (a *local* or *gauge* symmetry)? Then $\partial_\mu\psi \to \mathrm{e}^{\mathrm{i}\alpha}(\partial_\mu\psi + \mathrm{i}(\partial_\mu\alpha)\psi)$ and the kinetic term is no longer invariant: the extra piece is $\propto \partial_\mu\alpha$.

The fix is to introduce a new field $A_\mu(x)$ and replace $\partial_\mu \to D_\mu = \partial_\mu + \mathrm{i}qA_\mu$, with $A_\mu \to A_\mu - \frac{1}{q}\partial_\mu\alpha$ when $\psi \to \mathrm{e}^{\mathrm{i}\alpha}\psi$. Then $D_\mu\psi \to \mathrm{e}^{\mathrm{i}\alpha}D_\mu\psi$, and

$$\mathcal{L} = (D_\mu\psi)^*(D^\mu\psi) - m^2\psi^*\psi = \partial_\mu\psi^*\partial^\mu\psi - m^2\psi^*\psi \;-\; qA_\mu\,\mathrm{i}(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) + q^2A_\mu A^\mu\psi^*\psi. \quad (7.1)$$

The middle term is $-qA_\mu J^\mu$ with J^μ precisely the Noether current (6.4): the gauge field couples to the conserved current, which is how electromagnetic coupling $-A_\mu J^\mu$ arises. The new field A_μ needs its own gauge-invariant kinetic term, $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, which is Maxwell's theory. (Conventions for the sign of q and the relative sign in D_μ vary between textbooks.) The punchline: *demanding local $U(1)$ symmetry forces the existence of the photon and fixes its coupling to the Noether current.*

Bridge to QFT

This is the subject of L&B ch. 14 on gauge fields, and of the chapters on quantum electrodynamics that follow. The other half of the story—the global symmetry's Noether current is conserved, but the local symmetry's "current" is tied to the gauge field's equation of motion $\partial_\mu F^{\mu\nu} = qJ^\nu$ (up to sign conventions), whose divergence gives $\partial_\nu J^\nu = 0$ identically—is worth rereading when you get there.

8 Summary

Symmetry	**Transformation**	**Conserved quantity**
Particles		
Cyclic coordinate q_k	$\delta q_k = \varepsilon$	$p_k = \partial L/\partial\dot{q}_k$
Space translation along $\boldsymbol{n}$	$\delta\boldsymbol{r}_a = \varepsilon\boldsymbol{n}$	$\boldsymbol{n}\cdot\boldsymbol{P}$
Rotation about $\boldsymbol{n}$	$\delta\boldsymbol{r}_a = \varepsilon\,\boldsymbol{n}\times\boldsymbol{r}_a$	$\boldsymbol{n}\cdot\boldsymbol{J}$, $\boldsymbol{J} = \sum \boldsymbol{r}_a\times\boldsymbol{p}_a$
Time translation	$\delta q = \varepsilon\dot{q}$, $F = L$	$H = \sum p\dot{q} - L$
Galilean boost	$\delta\boldsymbol{r}_a = \varepsilon\boldsymbol{n}t$, $F = M\boldsymbol{n}\cdot\boldsymbol{R}$	$\boldsymbol{n}\cdot(\boldsymbol{P}t - M\boldsymbol{R})$
Fields		
$U(1)$ phase, $\psi \to \mathrm{e}^{-\mathrm{i}\alpha}\psi$	$\Delta\psi = -\mathrm{i}\psi$	$J^\mu = \mathrm{i}(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*)$, $Q = \int J^0$
Spacetime translation	$\Delta\phi = \partial_\nu\phi$	$T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial^\nu\phi - g^{\mu\nu}\mathcal{L}$; $P^\nu = \int T^{0\nu}$
Lorentz rotations	—	J^{ij} (angular momentum), boost charges K^i

Table 1: Symmetry ↔ conserved quantity.

Know cold

- Definition: $\delta q = \varepsilon K$ is a symmetry if $\delta L = \varepsilon\,\mathrm{d}F/\mathrm{d}t$; then $Q = \sum_i p_i K_i - F$ is conserved. Be able to reproduce the three-line proof.
- $\delta\mathcal{L} = \varepsilon\partial_\mu W^\mu \Rightarrow J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\Delta\phi - W^\mu$, $\partial_\mu J^\mu = 0$, $Q = \int J^0$ constant (given falloff at infinity). Both require the equations of motion.
- Translation $\to \boldsymbol{p}$; rotation $\to \boldsymbol{r}\times\boldsymbol{p}$; time translation (with $F = L$) $\to H$; boost has $F \neq 0$.
- Charges generate: $\delta q = \varepsilon\{q, Q\}$; $\{Q, H\} = 0$. Quantum: $[\hat{Q}, \hat{H}] = 0$, $U = \mathrm{e}^{\mathrm{i}\varepsilon\hat{Q}}$ unitary.
- $U(1)$ current $J^\mu = \mathrm{i}(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*)$ (sign conventional); positive-frequency modes carry $Q > 0$.
- $T^{\mu\nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial^\nu\phi - g^{\mu\nu}\mathcal{L}$; $T^{00} = \mathcal{H}$; $T^{0i} = -\pi\partial_i\phi$.
- Gauging $U(1)$ introduces A_μ with $D_\mu = \partial_\mu + \mathrm{i}qA_\mu$ and coupling $-qA_\mu J^\mu$.

9 Exercises

Solutions are collected in Section 10. Try each before looking.

Exercise 1 *(Uniform gravity).* A particle of mass m moves in the potential $V = mgz$. Write L. Which components of linear momentum are conserved? Is angular momentum conserved about any axis? Identify the symmetry associated with each conserved quantity, and find also the conserved quantity associated with a boost in the x-direction.

Exercise 2 *(Screw symmetry).* In cylindrical coordinates (ρ, φ, z) a particle of mass m has Lagrangian $L = \frac{1}{2}m(\dot\rho^2 + \rho^2\dot\varphi^2 + \dot z^2) - V(\rho,\, z - c\varphi)$, with c a constant (a helical potential; $2\pi c$ is the pitch). Identify the symmetry (a rotation combined with a translation), find K, and compute the Noether charge. Then repeat for $V(\rho, \varphi - kz)$ and compare the results (which combination of L_z and p_z is conserved in each case?). Check the signs by differentiating the charge directly.

Exercise 3 *(Dilatation of an inverse-square potential).* A particle in three dimensions has $L = \frac{1}{2}m\dot{\boldsymbol{r}}^2 - \lambda/r^2$. Consider the scaling $\boldsymbol{r}'(t) = \mu\boldsymbol{r}(t/\mu^2)$. Take $\mu = 1+\varepsilon$ and show that $\delta\boldsymbol{r} = \varepsilon(\boldsymbol{r} - 2t\dot{\boldsymbol{r}})$, that $\delta L = -2\varepsilon\,\mathrm{d}(tL)/\mathrm{d}t$, and hence that $D = \boldsymbol{p}\cdot\boldsymbol{r} - 2tH$ is conserved. Verify by direct differentiation. (For the Kepler potential $\propto 1/r$ the same scaling maps solutions to solutions but does *not* give a conserved charge as δL is not a total derivative; why?)

Exercise 4 *(Angular momentum brackets).* With $\{x_i, p_j\} = \delta_{ij}$ show that $\{L_z, x\} = y$, $\{L_z, y\} = -x$, $\{L_z, z\} = 0$, and the same for p. Hence show $\{L_z, r^2\} = 0$ and $\{L_z, \boldsymbol{p}^2\} = 0$ and conclude that L_z is conserved for $H = \boldsymbol{p}^2/2m + V(r)$. Finally show $\{L_x, L_y\} = L_z$.

Exercise 5 *($SO(2)$ model and the complex field).* Two real fields have $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi_1)^2 + \frac{1}{2}(\partial_\mu\phi_2)^2 - \frac{1}{2}m^2(\phi_1^2 + \phi_2^2)$. Show this is invariant under the rotation $\phi_1 \to \phi_1 + \varepsilon\phi_2$, $\phi_2 \to \phi_2 - \varepsilon\phi_1$ and compute the Noether current. Show it equals (6.4) when $\psi = (\phi_1 + \mathrm{i}\phi_2)/\sqrt{2}$, and that $\mathcal{L}$ equals (6.1).

Exercise 6 *(Massless scalar: $T^{\mu\nu}$ and its trace).* For $\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi$ in $3+1$ dimensions, compute $T^{\mu\nu}$, check $\partial_\mu T^{\mu\nu} = 0$ on-shell, and compute the trace $T^\mu{}_\mu$. Show it is nonzero. Show that the "improved" tensor $\tilde{T}^{\mu\nu} = T^{\mu\nu} - \frac{1}{6}(\partial^\mu\partial^\nu - g^{\mu\nu}\Box)\phi^2$ is still conserved, has the same P^ν, and is traceless on-shell.

Exercise 7 *(Boost in a uniform force field).* A particle moves in one dimension with $L = \frac{1}{2}m\dot{x}^2 + fx$, with f constant. Show that $\delta x = \varepsilon t$ is a symmetry with $F \neq 0$, find F, and compute Q. Verify that Q is conserved, and interpret.

Exercise 8 *(Continuity and charge in a volume).* Let $\partial_\mu J^\mu = 0$ and let $Q_V(t) = \int_V \mathrm{d}^3x\, J^0$ for a fixed region V with boundary S. Show that $\dot{Q}_V = -\oint_S \boldsymbol{J}\cdot\mathrm{d}\boldsymbol{S}$, and explain why this shows that charge can change in V only by flowing across S. Deduce $\dot{Q} = 0$ for $V = \mathbb{R}^3$.

Exercise 9 *(Explicit time dependence).* Show that $\mathrm{d}H/\mathrm{d}t = -\partial L/\partial t$ on-shell for $L(q,\dot{q},t)$. Explain in terms of Noether's theorem why H is not conserved for $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k(t)x^2$ with time-dependent k.

Exercise 10 *(Adding a total derivative).* Let $L' = L + \mathrm{d}G(q,t)/\mathrm{d}t$. Show that L' has the same symmetry $\delta q_i = \varepsilon K_i$ with $\delta L' = \varepsilon\,\mathrm{d}F'/\mathrm{d}t$ for $F' = F + \sum_i K_i\,\partial G/\partial q_i$ and that the Noether charge is the same: $Q' = Q$.

Exercise 11 *($U(1)$ charge generates phase rotations).* For the complex scalar field use the canonical Poisson brackets $\{\psi(\boldsymbol{x}),\,\pi(\boldsymbol{y})\} = \delta^3(\boldsymbol{x}-\boldsymbol{y})$, $\{\psi^*(\boldsymbol{x}),\,\pi^*(\boldsymbol{y})\} = \delta^3(\boldsymbol{x}-\boldsymbol{y})$ with all others vanishing. Write Q in terms of ψ,π and show $\{\psi,\,Q\} = -\mathrm{i}\psi$ and $\{\psi^*,\,Q\} = \mathrm{i}\psi^*$, so that $\delta\psi = \varepsilon\,\{\psi,\,Q\}$ reproduces $\Delta\psi = -\mathrm{i}\psi$.

10 Solutions

Solution to Exercise 1. $L = \frac{1}{2}m(\dot{x}^2+\dot{y}^2+\dot{z}^2) - mgz$. The coordinates x, y are cyclic, so $p_x = m\dot{x}$ and $p_y = m\dot{y}$ are conserved (translations in the horizontal plane). $p_z = m\dot{z}$ is *not* conserved ($\dot{p}_z = -mg$): L depends on z. Rotation about z ($\delta x = -\varepsilon y$, $\delta y = \varepsilon x$) leaves $\dot{x}^2+\dot{y}^2$ invariant and V is independent of x, y, so $L_z = m(x\dot{y} - y\dot{x})$ is conserved. Rotation about x or y changes z and hence V (not a total derivative), so L_x, L_y are not conserved: $\dot{\boldsymbol{L}} = \boldsymbol{r}\times\boldsymbol{F}$ with $\boldsymbol{F} = -mg\hat{z}$ has nonzero x, y components. The energy $E = \frac{1}{2}m\dot{\boldsymbol{r}}^2 + mgz$ is conserved (L has no explicit t). Boost in x: $\delta x = \varepsilon t$, $\delta\dot{x} = \varepsilon$, $\delta L = m\dot{x}\,\varepsilon = \varepsilon\,\mathrm{d}(mx)/\mathrm{d}t$, so $F = mx$ and $Q = p_x t - mx = m(\dot{x}t - x)$. Check: $\dot{Q} = m\ddot{x}t + m\dot{x} - m\dot{x} = 0$ since $\ddot{x} = 0$. A boost in z is also a symmetry, now with a t-dependent F: $\delta z = \varepsilon t$ gives $\delta L = \varepsilon(m\dot{z} - mgt) = \varepsilon\,\mathrm{d}(mz - \frac{1}{2}mgt^2)/\mathrm{d}t$, so $Q = m\dot{z}t - mz + \frac{1}{2}mgt^2$, and $\dot{Q} = t(m\ddot{z} + mg) = 0$ ✓. (This is the 3D version of Exercise 7.)

Solution to Exercise 2. Take $\varphi \to \varphi + \varepsilon$ and $z \to z + c\varepsilon$ (so $z - c\varphi$ is unchanged; ρ unchanged). Thus $K_\varphi = 1$, $K_z = c$. $\dot{\varphi}, \dot{z}, \rho$ are unchanged, so $\delta L = 0$, $F = 0$. Then

$$Q = p_\varphi K_\varphi + p_z K_z = m\rho^2\dot{\varphi} + c\,m\dot{z} = L_z + c\,p_z. \tag{10.1}$$

Direct check: $\dot{p}_\varphi = \partial L/\partial\varphi = -\partial_2 V\cdot(-c) = c\,\partial_2 V$, where $\partial_2 V$ denotes derivative with respect to the second argument $u = z - c\varphi$ (since $\partial u/\partial\varphi = -c$ and $\partial L/\partial\varphi = -V_u\cdot(-c) = +cV_u$); and $\dot{p}_z = \partial L/\partial z = -V_u$. So $\dot{p}_\varphi + c\,\dot{p}_z = cV_u - cV_u = 0$ ✓. For $V(\rho, \varphi - kz)$ the invariant combination $v = \varphi - kz$ requires $\delta\varphi = k\,\delta z$: $K_z = 1, K_\varphi = k$, giving $Q = p_z + k\,L_z$. Check: $\dot{p}_\varphi = -V_v$, $\dot{p}_z = +kV_v$, so $\dot{p}_z + k\dot{p}_\varphi = kV_v - kV_v = 0$ ✓. The two are consistent: $\varphi - kz = -k(z - c\varphi)$ with $c = 1/k$, and $p_z + kL_z = k(L_z + c\,p_z)$. So "$L_z + c\,p_z$" goes with $V(\rho, z - c\varphi)$, whereas $V(\rho, \varphi - kz)$ gives $p_z + kL_z$ (not $L_z + kp_z$). General rule: the coefficient multiplying p_z relative to L_z is the amount of z-translation accompanying a unit rotation.

Solution to Exercise 3. $\boldsymbol{r}'(t) = \mu\boldsymbol{r}(t/\mu^2)$. With $\mu = 1+\varepsilon$: $\boldsymbol{r}'(t) = (1+\varepsilon)\boldsymbol{r}(t(1-2\varepsilon)) = \boldsymbol{r} + \varepsilon(\boldsymbol{r} - 2t\dot{\boldsymbol{r}})$, so $K = \boldsymbol{r} - 2t\dot{\boldsymbol{r}}$. Then $\delta\dot{\boldsymbol{r}} = \varepsilon(\dot{\boldsymbol{r}} - 2\dot{\boldsymbol{r}} - 2t\ddot{\boldsymbol{r}}) = -\varepsilon(\dot{\boldsymbol{r}} + 2t\ddot{\boldsymbol{r}})$. With $T = \frac{1}{2}m\dot{\boldsymbol{r}}^2$:

$$\delta T = m\dot{\boldsymbol{r}}\cdot\delta\dot{\boldsymbol{r}} = -\varepsilon\big(m\dot{\boldsymbol{r}}^2 + 2tm\dot{\boldsymbol{r}}\cdot\ddot{\boldsymbol{r}}\big) = -\varepsilon\Big(2T + 2t\frac{\mathrm{d}T}{\mathrm{d}t}\Big). \tag{10.2}$$

For $V = \lambda r^{-2}$, $\delta(r^2) = 2\boldsymbol{r}\cdot\delta\boldsymbol{r} = 2\varepsilon(r^2 - 2t\boldsymbol{r}\cdot\dot{\boldsymbol{r}}) = 2\varepsilon\,r^2 + 2\varepsilon t\cdot(-2\boldsymbol{r}\cdot\dot{\boldsymbol{r}})$. Since $\delta(r^{-2}) = -r^{-4}\delta(r^2)$ and $\mathrm{d}(r^{-2})/\mathrm{d}t = -2r^{-4}\boldsymbol{r}\cdot\dot{\boldsymbol{r}}$:

$$\delta V = \lambda\,\delta(r^{-2}) = \varepsilon\lambda\big(-2r^{-2} + 4t\,r^{-4}\boldsymbol{r}\cdot\dot{\boldsymbol{r}}\big) = \varepsilon\Big(-2V - 2t\frac{\mathrm{d}V}{\mathrm{d}t}\Big). \tag{10.3}$$

Therefore $\delta L = \delta T - \delta V = \varepsilon\big(-2(T-V) - 2t\,\mathrm{d}(T-V)/\mathrm{d}t\big) = -2\varepsilon(L + t\dot{L}) = -2\varepsilon\,\mathrm{d}(tL)/\mathrm{d}t$: $F = -2tL$. The charge is

$$Q = \boldsymbol{p}\cdot K - F = \boldsymbol{p}\cdot\boldsymbol{r} - 2t\,\boldsymbol{p}\cdot\dot{\boldsymbol{r}} + 2tL = \boldsymbol{p}\cdot\boldsymbol{r} - 2tH = D. \tag{10.4}$$

Direct check: $\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{p}\cdot\boldsymbol{r}) = \dot{\boldsymbol{p}}\cdot\boldsymbol{r} + \boldsymbol{p}\cdot\dot{\boldsymbol{r}} = -\boldsymbol{r}\cdot\nabla V + 2T$. For $V \propto r^{-2}$, Euler's theorem gives $\boldsymbol{r}\cdot\nabla V = -2V$, so this is $2V + 2T = 2E$. And $\frac{\mathrm{d}}{\mathrm{d}t}(2tH) = 2E$ since H is conserved. Hence $\dot{D} = 0$ ✓. For $V = -k/r$: $\boldsymbol{r}\cdot\nabla V = -V$, $\frac{\mathrm{d}(\boldsymbol{p}\cdot\boldsymbol{r})}{\mathrm{d}t} = 2T + V \neq 2E$; the scaling $\boldsymbol{r}\to\mu\boldsymbol{r}$, $t\to\mu^{3/2}t$ maps solutions to solutions (Kepler's third law $T^2\propto a^3$) but it rescales the action by a constant factor $\mu^{1/2}$ rather than leaving it invariant, so it is not a symmetry of the action in the sense of (3.3) and no Noether charge results. (A symmetry must have δL a total derivative with the action unchanged, not merely scaled.)

Solution to Exercise 4. $L_z = xp_y - yp_x$. Using $\{x_i,\,p_j\} = \delta_{ij}$ and Leibniz: $\{L_z,\,x\} = \{xp_y,\,x\} - \{yp_x,\,x\} = 0 - y\,\{p_x,\,x\} = -y(-1) = y$. $\{L_z,\,y\} = x\,\{p_y,\,y\} = -x$. $\{L_z,\,z\} = 0$. For momenta: $\{L_z,\,p_x\} = \{xp_y,\,p_x\} - \{yp_x,\,p_x\} = p_y\,\{x,\,p_x\} = p_y$; $\{L_z,\,p_y\} = -p_x\,\{y,\,p_y\} = -p_x$; $\{L_z,\,p_z\} = 0$. So $\{L_z,\,\cdot\}$ acts on (x,y) and (p_x,p_y) as $(\cdot)\mapsto(+y,-x)$ and $(+p_y,-p_x)$. Since $\{x,\,L_z\} = -y$, $\{y,\,L_z\} = x$, $\delta\boldsymbol{r} = \varepsilon\,\{\boldsymbol{r},\,L_z\} = \varepsilon(-y,x,0) = \varepsilon\hat{z}\times\boldsymbol{r}$ ✓(rotation), and $\delta\boldsymbol{p} = \varepsilon\,\{\boldsymbol{p},\,L_z\} = \varepsilon(-p_y,p_x,0) = \varepsilon\hat{z}\times\boldsymbol{p}$. Then $\{L_z,\,r^2\} = 2x\,y + 2y(-x) = 0$ and $\{L_z,\,\boldsymbol{p}^2\} = 2p_xp_y + 2p_y(-p_x) = 0$. For $H = \boldsymbol{p}^2/2m + V(r)$, $\{L_z,\,V(r)\} = V'(r)\,\{L_z,\,r\} = 0$ since $\{L_z,\,r^2\} = 0$. So $\{L_z,\,H\} = 0$ and $\dot{L}_z = \{L_z,\,H\} = 0$ ✓. Finally $L_x = yp_z - zp_y$, $L_y = zp_x - xp_z$: the only nonvanishing brackets are $\{yp_z,\,zp_x\} = y\,\{p_z,\,z\}\,p_x = -yp_x$ and $\{-zp_y,\,-xp_z\} = \{zp_y,\,xp_z\} = x\,\{z,\,p_z\}\,p_y = xp_y$ (the cross-pairs $\{yp_z,\,-xp_z\}$ and $\{-zp_y,\,zp_x\}$ vanish). Total: $xp_y - yp_x = L_z$ ✓.

Solution to Exercise 5. Under $\delta\phi_1 = \varepsilon\phi_2$, $\delta\phi_2 = -\varepsilon\phi_1$: $\delta(\phi_1^2 + \phi_2^2) = 2\varepsilon(\phi_1\phi_2 - \phi_2\phi_1) = 0$ and likewise $\delta[(\partial\phi_1)^2 + (\partial\phi_2)^2] = 2\varepsilon(\partial\phi_1\cdot\partial\phi_2 - \partial\phi_2\cdot\partial\phi_1) = 0$. So $\delta\mathcal{L} = 0$, $W^\mu = 0$, $\Delta\phi_1 = \phi_2$, $\Delta\phi_2 = -\phi_1$. With $\partial\mathcal{L}/\partial(\partial_\mu\phi_a) = \partial^\mu\phi_a$:

$$J^\mu = \partial^\mu\phi_1\,\phi_2 + \partial^\mu\phi_2\,(-\phi_1) = \phi_2\partial^\mu\phi_1 - \phi_1\partial^\mu\phi_2. \tag{10.5}$$

Complex field: $\psi = (\phi_1 + \mathrm{i}\phi_2)/\sqrt{2}$. Then $\partial\psi^* \cdot \partial\psi = \frac{1}{2}(\partial\phi_1 - \mathrm{i}\partial\phi_2)\cdot(\partial\phi_1 + \mathrm{i}\partial\phi_2) = \frac{1}{2}[(\partial\phi_1)^2 + (\partial\phi_2)^2]$ and $\psi^*\psi = \frac{1}{2}(\phi_1^2 + \phi_2^2)$, so $\mathcal{L}$ coincides with (6.1). Also $\delta\psi = \frac{1}{\sqrt{2}}(\varepsilon\phi_2 - \mathrm{i}\varepsilon\phi_1) = -\mathrm{i}\varepsilon\psi$ ✓, matching $\Delta\psi = -\mathrm{i}\psi$. The current (6.4):

$$\begin{aligned}\mathrm{i}(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) &= \tfrac{\mathrm{i}}{2}\big[(\phi_1 - \mathrm{i}\phi_2)(\partial^\mu\phi_1 + \mathrm{i}\partial^\mu\phi_2) \\ &\qquad - (\phi_1 + \mathrm{i}\phi_2)(\partial^\mu\phi_1 - \mathrm{i}\partial^\mu\phi_2)\big] \\ &= \tfrac{\mathrm{i}}{2}\big[2\mathrm{i}\,\phi_1\partial^\mu\phi_2 - 2\mathrm{i}\,\phi_2\partial^\mu\phi_1\big] \\ &= \phi_2\partial^\mu\phi_1 - \phi_1\partial^\mu\phi_2,\end{aligned}$$

agreeing with the $SO(2)$ current ✓.

Solution to Exercise 6. $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$, so $T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \frac{1}{2}g^{\mu\nu}\partial_\lambda\phi\partial^\lambda\phi$. Divergence: $\partial_\mu T^{\mu\nu} = \Box\phi\,\partial^\nu\phi + \partial^\mu\phi\,\partial_\mu\partial^\nu\phi - \partial_\lambda\phi\,\partial^\nu\partial^\lambda\phi = \Box\phi\,\partial^\nu\phi = 0$ on-shell ($\Box\phi = 0$) ✓. Trace, with $g^\mu{}_\mu = 4$: $T^\mu{}_\mu = \partial_\mu\phi\partial^\mu\phi - 2\,\partial_\lambda\phi\partial^\lambda\phi = -\partial_\lambda\phi\,\partial^\lambda\phi \neq 0$. (In d spacetime dimensions the trace is $(1 - d/2)(\partial\phi)^2$, vanishing only for $d = 2$.) Improved tensor: $\tilde{T}^{\mu\nu} = T^{\mu\nu} - \frac{1}{6}(\partial^\mu\partial^\nu - g^{\mu\nu}\Box)\phi^2$. The added term $X^{\mu\nu} = (\partial^\mu\partial^\nu - g^{\mu\nu}\Box)\phi^2$ satisfies $\partial_\mu X^{\mu\nu} = \partial^\nu\Box\phi^2 - \partial^\nu\Box\phi^2 = 0$ identically, so conservation is preserved. The change in P^ν is $\int \mathrm{d}^3x\, X^{0\nu}$: for $\nu = 0$, $X^{00} = (\partial_0^2 - \Box)\phi^2 = \nabla^2\phi^2$, a spatial divergence which integrates to zero; for $\nu = i$, $X^{0i} = \partial^0\partial^i\phi^2 = -\partial_t\partial_i\phi^2$, also a spatial derivative, so zero. Trace: $X^\mu{}_\mu = (\Box - 4\Box)\phi^2 = -3\Box\phi^2$, so $\tilde{T}^\mu{}_\mu = -(\partial\phi)^2 + \frac{1}{2}\Box\phi^2$. Now $\Box\phi^2 = 2\partial_\mu\phi\partial^\mu\phi + 2\phi\Box\phi = 2(\partial\phi)^2$ on-shell. Hence $\tilde{T}^\mu{}_\mu = -(\partial\phi)^2 + (\partial\phi)^2 = 0$ ✓. The tracelessness reflects the classical scale (conformal) invariance of the massless scalar in $3+1$ dimensions.

Solution to Exercise 7. $\delta x = \varepsilon t$, $\delta\dot{x} = \varepsilon$. $\delta L = m\dot{x}\,\varepsilon + f\,\varepsilon t = \varepsilon\frac{\mathrm{d}}{\mathrm{d}t}(mx + \frac{1}{2}ft^2)$, so $F = mx + \frac{1}{2}ft^2$. Charge: $Q = p\,K - F = m\dot{x}\,t - mx - \frac{1}{2}ft^2$. Check: $\dot{Q} = m\ddot{x}\,t + m\dot{x} - m\dot{x} - ft = t(m\ddot{x} - f) = 0$ since the equation of motion is $m\ddot{x} = f$ ✓. Interpretation: explicitly $x(t) = x_0 + v_0 t + \frac{f}{2m}t^2$, $\dot{x} = v_0 + ft/m$, so $Q = m(v_0 + ft/m)t - mx_0 - mv_0 t - \frac{1}{2}ft^2 - \frac{1}{2}ft^2 = -mx_0$: Q is minus m times the position at $t = 0$, which labels the trajectory. As in the free case, the boost charge says that the "centre" follows a prescribed motion.

Solution to Exercise 8. Integrate $\partial_t J^0 = -\nabla\cdot\boldsymbol{J}$ over the fixed region V: $\dot{Q}_V = \int_V \partial_t J^0\,\mathrm{d}^3x = -\int_V \nabla\cdot\boldsymbol{J}\,\mathrm{d}^3x = -\oint_S \boldsymbol{J}\cdot\mathrm{d}\boldsymbol{S}$ by the divergence theorem. The right-hand side is the net flux of $\boldsymbol{J}$ out of V, so Q_V decreases exactly when charge leaves through S: there is no way for Q_V to change other than by flow across the boundary (local conservation). For $V = \mathbb{R}^3$, S is at infinity, where $\boldsymbol{J} \to 0$ fast enough ($|\boldsymbol{J}|r^2 \to 0$), so the flux vanishes and $\dot{Q} = 0$. Note that local conservation is stronger than global conservation: a global Q could in principle stay constant while charge disappears here and reappears elsewhere; $\partial_\mu J^\mu = 0$ forbids that. It is also what is required by relativity, since "simultaneously" is observer-dependent.

Solution to Exercise 9. $H = \sum p_i\dot{q}_i - L$. Then $\dot{H} = \sum(\dot{p}_i\dot{q}_i + p_i\ddot{q}_i) - \sum\left(\partial_{q_i}L\,\dot{q}_i + \partial_{\dot{q}_i}L\,\ddot{q}_i\right) - \partial_t L = \sum(\dot{p}_i - \partial_{q_i}L)\dot{q}_i - \partial_t L = -\partial_t L$ on-shell (since $p_i = \partial_{\dot{q}_i}L$ cancels the $\ddot{q}$ terms and $\dot{p}_i = \partial_{q_i}L$). For $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k(t)x^2$: $\dot{H} = \frac{1}{2}\dot{k}x^2 \neq 0$. Noether: under $x \to x + \varepsilon\dot{x}$, $\delta L = \varepsilon(\mathrm{d}L/\mathrm{d}t - \partial_t L) = \varepsilon\,\mathrm{d}L/\mathrm{d}t + \frac{1}{2}\varepsilon\dot{k}x^2$, and the extra term $\frac{1}{2}\dot{k}x^2$ is not a total derivative of a function of q, t (for arbitrary $k(t)$), so the transformation is not a symmetry in the sense of (3.3). The charge H then fails to be conserved by exactly $\varepsilon\,\partial_t L$. If k is constant, the extra term vanishes and H is conserved ✓.

Solution to Exercise 10. $\delta\dfrac{\mathrm{d}G}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\delta G)$ and $\delta G = \sum_i \varepsilon K_i\,\partial G/\partial q_i$ since $G = G(q,t)$ and t is not varied. So $\delta L' = \varepsilon\,\mathrm{d}F/\mathrm{d}t + \varepsilon\frac{\mathrm{d}}{\mathrm{d}t}\big(\sum_i K_i\partial_{q_i}G\big)$, i.e. $F' = F + \sum_i K_i\partial_{q_i}G$. The momenta change:

$p'_i = \partial L'/\partial\dot{q}_i = p_i + \partial G/\partial q_i$, since $\mathrm{d}G/\mathrm{d}t = \sum_j \dot{q}_j \partial_{q_j} G + \partial_t G$. Therefore $Q' = \sum_i p'_i K_i - F' = \sum_i p_i K_i + \sum_i K_i \partial_{q_i} G - F - \sum_i K_i \partial_{q_i} G = Q$ ✓. Moral: the charge—a physical, measurable quantity—is unaffected by the total-derivative ambiguity of L, although p_i and F individually shift. (The Hamiltonian-language version is the canonical transformation $p \to p + \partial G/\partial q$.)

Solution to Exercise 11. $\pi = \partial\mathcal{L}/\partial\dot{\psi} = \dot{\psi}^*$ and $\pi^* = \dot{\psi}$. Then $J^0 = \mathrm{i}(\psi^*\dot{\psi} - \psi\dot{\psi}^*) = \mathrm{i}(\psi^*\pi^* - \psi\pi)$, so

$$Q = \mathrm{i}\int \mathrm{d}^3y\,(\psi^*(\boldsymbol{y})\pi^*(\boldsymbol{y}) - \psi(\boldsymbol{y})\pi(\boldsymbol{y})). \tag{10.6}$$

Now the only nonvanishing bracket of $\psi(\boldsymbol{x})$ is with π. So

$$\{\psi(\boldsymbol{x}),\, Q\} = -\mathrm{i}\int \mathrm{d}^3y\,\psi(\boldsymbol{y})\;\{\psi(\boldsymbol{x}),\,\pi(\boldsymbol{y})\} = -\mathrm{i}\psi(\boldsymbol{x}).$$

Similarly, ψ^* only has nonvanishing bracket with π^*, giving $\{\psi^*(\boldsymbol{x}),\, Q\} = \mathrm{i}\psi^*(\boldsymbol{x})$ by the same steps. Hence $\delta\psi = \varepsilon\,\{\psi,\, Q\} = -\mathrm{i}\varepsilon\psi$, the infinitesimal form of $\psi \to \mathrm{e}^{-\mathrm{i}\varepsilon}\psi$ ✓. The field-theory version of "the charge generates its symmetry"; under quantization $\{\cdot,\,\cdot\} \to -\mathrm{i}[\cdot,\cdot]$, giving $[\hat{Q}, \hat{\psi}] = -\hat{\psi}$, so $\hat{\psi}$ lowers the charge by one—the statement that $\hat{\psi}$ annihilates a particle (or creates an antiparticle).

Suggested further reading

- L. Susskind and G. Hrabovsky, *The Theoretical Minimum: What You Need to Know to Start Doing Physics* (Classical Mechanics), Lecture 7 (symmetries and conservation laws) and Lecture 10 (Poisson brackets and symmetries).
- H. Goldstein, C. Poole, J. Safko, *Classical Mechanics*, §2.7 (conservation theorems and symmetry properties) and ch. 13 (classical field theory, including Noether's theorem for fields).
- D. Tong, *Quantum Field Theory* (Cambridge lecture notes), Section 1 (classical field theory, including Noether's theorem and the energy–momentum tensor).
- D. Tong, *Classical Dynamics* (Cambridge lecture notes), for the Lagrangian and Hamiltonian background.